Tide retrospective: the contradiction, packaged

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The germbij arc's quantitative chains end in lower bounds of the form $\kappa\, t^{1-m-d/2} \le \Delta(t)$, and the note's Theorem 7.3 derives its conclusion by observing that such a bound contradicts $\Delta(t) = o(t^{-\infty})$. That last inference lived only in prose. This tide, the last item on the arc's recorded follow-up list, packages it in Lean, so the theorem's conclusion (not merely its quantitative core) is a formal statement.

2 The thing formalised

Two declarations in `Laplace/Decay.lean`, sorry-free with zero warnings. The general lemma[1]: a function Δ with $\kappa\, t^{\gamma} \le \Delta(t)$ for $t \ge T_0$ ($\kappa > 0$, γ any real) cannot satisfy $\Delta = o(t^{-N})$ for every $N \in \mathbb{N}$. The corollary[2]: under the analytic multivariate hypotheses (power series for $L_2 - L_1$ with vanishing lower diagonal terms, a sphere point where the leading diagonal term is nonzero, nonnegativity, quadratic domination, bump), the pencil difference

$$t \;\longmapsto\; \int (L_2 - L_1)\, \psi \left(e^{-tL_1} - e^{-tL_2}\right) dw$$

is *not* $o(t^{-N})$ for all N. Contrapositively: if the two Laplace families agree beyond all orders against the single observable $(L_2 - L_1)\psi$, no such power series data can exist, which is the germbij Theorem 7.3 contradiction verbatim. What remains outside Lean is only the note's reduction to this observable: choosing the point where the germ of g is nonzero and the bump ψ (a compactness argument), and quantifying over all test functions.

3 Proof strategy

Archimedes gives $N > -\gamma$; the little-o definition at $\varepsilon = \kappa/2$ gives an eventual upper bound $|\Delta(t)| \le (\kappa/2)\, t^{-N}$; for $t \ge \max(T_0, 1)$ the lower bound gives $\kappa\, t^{-N} \le \kappa\, t^{\gamma} \le \Delta(t)$ since $t^{\gamma+N} \ge 1$; cancelling the positive factor t^{-N} leaves $\kappa \le \kappa/2$. The corollary destructures the analytic lower bound and rewrites $t \cdot t^{-m-d/2}$ as $t^{1-m-d/2}$. The consult confirmed the statement (including that $\forall N \in \mathbb{N}$ is the standard rendering of superpolynomial decay, equivalent to real exponents by monotonicity) and the idiom set; both parties voted for the two-declaration file.

[1] `lower_bound_not_superpolynomial`.
[2] `analytic_pencil_difference_not_superpolynomial`.

4 Roadblocks and resolutions

Two elaboration-level fixes. `mul_le_mul_right` did not resolve to the ordered-field iff form in this context, so the cancellation became `le_of_mul_le_mul_right`. And in the corollary's `refine`, the lower-bound threshold T_0 occurs only inside the hole, so Lean cannot infer it; it is passed explicitly with the binder annotated. Both are one-line repairs of a familiar class.

5 What was Mathlib, what was new

All Mathlib: the little-o characterisation, `rpow` monotonicity and addition, Archimedes, and ordered-field cancellation. New: nothing mathematically; the value is that the arc's headline now terminates in a negation a reader can quote, rather than in an inequality plus a prose remark.

6 Lessons

Lemmas whose name is shared between an iff form and a mul-mono form (`mul_le_mul_right` and kin) are worth checking with a `#check` before use; the failure mode (resolution to the wrong namespace) produces a confusing "field `mp` does not exist" rather than a name error. When a `refine` hole is the only occurrence of an implicit variable, pass it explicitly at the call.

7 Follow-ups

- Tag the note's Theorem 7.3 with the corollary once merged; it is now the closest Lean statement to the theorem as written.
- The arc is complete; no further Lean follow-ups are recorded. The remaining softening candidates (an `iteratedFDeriv` interface, the lean-common upstreaming of the cylinder majorant) are maintenance, not tides.